

Addressing econometric issues on how to construct theoretical based exchange rate misalignment estimates: searching for a comprehensive approach.

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Talking points

- 1 Motivation and Goals
- 2 Approaches to address RER misalignment
- 3 Addressing Important Econometric Issues
- 4 Empirical Illustrations
- 5 Possible Extensions

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Exchange Rate Misalignment definition and relevance:

- Effective exchange rate misalignment:
 - Deviations from a long run macroeconomic equilibrium
 - Economic discussion to define this equilibrium;
 - Empirical questions about how to estimate this equilibrium;
- Exchange rate misalignment can:
 - be related to financial crisis;
 - permanently influence trade flows;
 - a sign of serious macroeconomic imbalances;

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- Highlight the importance of jointly modeling RER, trade balance and net foreign asset and not only RER and NFA;
- How to test and decompose misalignment into economically meaningful parts;
 - Testing Long Run implications of economic based restrictions [2] to our problem;
- To propose a methodology that clearly highlight the gains of expanding information set;
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Basic Equilibrium Equation for RER.

- From many intertemporal macroeconomic models it's possible to obtain the following steady state equations

$$\overline{RER} = -\phi \overline{tb} + \lambda \overline{X} \quad (1)$$

- equation (1) shows that if a country can run a trade deficit in equilibrium the RER has to appreciate ;
- The term X accounts for any other factor affecting equilibrium Real Effective Exchange Rate (RER) such as Balassa-Samuelson effect, terms of trade, government consumption, trade and financial openness among others;

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- equation (2) shows that a country can run a trade deficits if revenues from NFA are large enough;

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A brief description of traditional BEER approach

- Fundamental Real exchange rate is calculated using an econometric model (based on (2)):
Applications: Faruquee [4], Clark [3], Ubide [9] and Kubota [6]
- Cointegration techniques based on selected series associated with fundamentals ;
- Econometric Questions:
- Detect long run relationship - cointegration analysis;
 - Different set of fundamentals can be tested;
- Perform permanent and transitory decomposition to better understand the adjustment towards equilibrium;
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Our starting point

- We assume for simplicity but test that DGP for the variables trade balance, real effective exchange rate and net foreign asset position and fundamentals is given by the following vector autoregressive model:

$$\Delta Y_t = \Gamma_1 \Delta Y_{t-1} + \dots + \Gamma_{k-1} \Delta Y_{t-k+1} + \alpha \beta' Y_{t-1} + \mu + \varepsilon_t \quad (3)$$

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Transitory and Permanent components

- Using the parameters from (3), it is possible to calculate the transitory (T_{it}) and permanent (P_{it}) components from the following equations:

$$P_t = \beta_{\perp}(\alpha'_{\perp}\beta_{\perp})^{-1}\alpha'_{\perp}Y_t \quad (4)$$

$$T_t = \alpha(\beta'\alpha)^{-1}\beta'Y_t \quad (5)$$

- Assuming that the real exchange rate is in the first position of the vector, and using the value of the error correction mechanism centered on their own means, we can calculate the misalignment using the following equation, then:

$$mis_t \equiv F_1(\beta'_1 Y_t - E([\beta'_1 Y_t])) + F_2(\beta'_2 Y_t - E([\beta'_2 Y_t]))$$

where $[V_1 \quad V_2] = (\beta'\alpha)^{-1}$ and V_i has dimension 2×1

$$F \equiv [F_1 F_2] = \begin{bmatrix} 1 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} \alpha_1 & \alpha_2 \end{bmatrix} V_1 \quad \begin{bmatrix} 1 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} \alpha_1 & \alpha_2 \end{bmatrix} V_2$$

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Are theoretical based restrictions valid? Test

• How to check identification restrictions of the theoretical model?

- Assume that the following sets of variable are modeled:

$$Y_t = [rer \quad tb \quad nfa \quad Fund1 \quad Fund2]'$$

where *Fund1* and *Fund2* are any variables related to fundamentals.

- If over-identifying restrictions hold, then the cointegrated vectors satisfy (7):

$$\beta_{r'} = \begin{bmatrix} 1 & 0 & b_{31} & b_{41} & b_{51} \\ 0 & 1 & b_{32} & 0 & 0 \end{bmatrix}'$$

where $\gamma = \{b_{31}, b_{32}, b_{41}, b_{51}\}$ are the coefficients of the long run relationship to be estimated.

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Does RER have a drift?

- **How to test that RER is not drifting?**

- (3) should satisfy:

$$\mu = \alpha\varphi \quad (8)$$

where φ is a matrix of order $r \times 1$ and r is the number of cointegrated relationship in the system.

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Adapting Blinder-Oaxaca decomposition:

- **Blinder-Oaxaca decomposition ([2, 8]).**

$$\begin{aligned}
 mis_{i,t}^{JMA} - mis_{i,t}^{BEER} = & \underbrace{\left(\frac{h_1^{JMA}}{h_1^{BEER}} - 1 \right) (h_1^{BEER} f_{1,t}^{BEER})}_{\text{Less weight on what we knew}} + \\
 & \underbrace{h_1^{JMA} (f_{1,t}^{JMA} - f_{1,t}^{BEER}) + h_2^{JMA} f_{2,t}^{JMA}}_{\text{What we learn from new model}}
 \end{aligned} \tag{9}$$

Adapting Blinder-Oaxaca decomposition:

- If cointegrated relationships are properly identified:

$$T^{-1}(f_{1,t}^{JMA} - f_{1,t}^{BEER}) \xrightarrow{P} 0 \quad (10)$$

- Both models are nested and it is interesting to investigate whether

- $\frac{h_1^{JMA}}{h_1^{BEER}} = 1$
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Conditions to guarantee $h_2^{JMA} = 0$

- Weight of GG decomposition are given by:

$$\begin{bmatrix} h_1^{JMA} & h_2^{JMA} \end{bmatrix} = \alpha(\beta' \alpha)^{-1} = \begin{bmatrix} \alpha_{11} & \alpha_{21} \\ \alpha_{12} & \alpha_{22} \end{bmatrix} \begin{pmatrix} \beta'_1 \alpha_1 & \beta'_1 \alpha_2 \\ \beta'_2 \alpha_1 & \beta'_2 \alpha_2 \end{pmatrix}^{-1} \quad (11)$$

- If $\beta'_1 \alpha_2 = 0$ and $\alpha_{21} = 0$ then $h_2^{JMA} = 0$.
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Example when $h_2^{JMA} = 0$

- Assume that:

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Example when $(\frac{h_1^{JMA}}{h_1^{BEER}} - 1) = 0$ and $h_2^{JMA} = 0$

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Relative price ratio	IMF	$BS_{i,t} = \frac{\frac{WSP_{i,t}}{IPCI_{i,t}}}{\sum_{j=1}^N w_j \frac{WSP_{j,t}}{CPI_{j,t}}}$

Cointegration Analysis: Trace

Countries	Rank	RER , NFA	RER, BSPIB	RER, BS	RER, TOT	RER, TB
Australia	r=0	***	**			*
	r=1					
Austria	r=0		**	**	n.a.	
	r=1				n.a.	
Brazil	r=0	**				**
	r=1					
China	r=0		***	***		**
	r=1		**			
Colombia	r=0					

Cointegration Analysis: Trace

Countries	Rank	RER , NFA	RER, BSPIB	RER, BS	RER, TOT	RER, TB
Denmark	r=0	*				
	r=1					
Finland	r=0	***		*		
	r=1					
Germany	r=0				*	***
	r=1					
Greece	r=0					
	r=1					
Ireland	r=0					

Cointegration Analysis: Trace

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Italy	r=0	*				
	r=1					
Japan	r=0	***	**		*	**
	r=1	*				
Korea	r=0	***	***			**
	r=1		***			
Mexico	r=0			*	***	***
	r=1				***	**
Netherlands	r=0		**	**		
	r=1					

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	r=1					
Norway	r=0	**				
	r=1					
Portugal	r=0				n.a.	
	r=1				n.a.	
Singapore	r=0	**	***			**
	r=1		**			*
Spain	r=0	***		*		*

Cointegration Analysis: Trace

Countries	Rank	RER , NFA	RER, BSPIB	RER, BS	RER, TOT	RER, TB
Sweden	$r=0$	**		**		
	$r=1$					
Switzerland	$r=0$		**	*	n.a.	
	$r=1$		*		n.a.	
Turkey	$r=0$	*				
	$r=1$					
United Kingdom	$r=0$	*				**
	$r=1$					*

Cointegration Analysis: Trace

Countries	Rank	RER , NFA	RER, BSPIB	RER, BS	RER, TOT	RER, TB
United States	$r=0$			*		
	$r=1$	*				
Uruguay	$r=0$					
	$r=1$					

Summary of evidence:

- Evidence of cointegration in pairs for most of countries between RER and Fundamentals.
- For those countries with no evidence in favor of cointegration in pairs there is evidence for trios and quartets (not reported but in the paper);

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Do TB and NFA cointegrate?

Countries	Rank	Eigenvalues	Trace	p-value	
Australia	r=0	0.345	18.596	0.3%	***
	r=1	0.000	0.012	95.1%	
Austria	r=0	0.111	8.011	23.7%	
	r=1	0.062	2.812	10.9%	
Brazil	r=0	0.264	13.729	2.8%	**
	r=1	0.006	0.258	68.4%	
China	r=0	0.139	9.137	16.2%	
	r=1	0.056	2.539	13.0%	
Colombia	r=0	0.150	7.795	25.5%	

Do TB and NFA cointegrate?

Countries	Rank	Eigenvalues	Trace	p-value	
Denmark	r=0	0.207	12.114	5.3%	*
	r=1	0.042	1.906	19.6%	
Finland	r=0	0.288	17.649	0.5%	***
	r=1	0.060	2.715	11.6%	
Germany	r=0	0.153	8.286	21.7%	
	r=1	0.022	0.991	37.2%	
Greece	r=0	0.174	10.263	10.8%	
	r=1	0.041	1.862	20.2%	
Ireland	r=0	0.129	6.453	38.4%	

Do TB and NFA cointegrate?

Countries	Rank	Eigenvalues	Trace	p-value	
Italy	r=0	0.214	10.686	9.2%	*
	r=1	0.002	0.080	84.0%	
Japan	r=0	0.273	17.607	0.5%	***
	r=1	0.078	3.578	6.8%	*
Korea	r=0	0.303	17.730	0.5%	***
	r=1	0.050	2.202	16.2%	
Mexico	r=0	0.195	9.527	14.1%	
	r=1	0.000	0.000	99.3%	
Netherland	r=0	0.057	2.682	87.2%	

Do TB and NFA cointegrate?

Countries	Rank	Eigenvalues	Trace	p-value	
New Zeland	r=0	0.244	12.036	5.5%	*
	r=1	0.000	0.014	94.4%	
Norway	r=0	0.268	13.828	2.7%	**
	r=1	0.003	0.128	78.9%	
Portugal	r=0	0.125	6.313	40.0%	
	r=1	0.017	0.729	45.4%	
Singapore	r=0	0.255	13.048	3.7%	**
	r=1	0.003	0.114	80.3%	
Spain	r=0	0.373	20.764	0.1%	***

Do TB and NFA cointegrate?

Countries	Rank	Eigenvalues	Trace	p-value	
Sweden	r=0	0.218	12.338	4.9%	**
	r=1	0.034	1.510	25.7%	
Switzerland	r=0	0.063	2.973	83.9%	
	r=1	0.002	0.109	80.8%	
Turkey	r=0	0.205	10.776	8.9%	*
	r=1	0.015	0.661	47.9%	
United Kingdom	r=0	0.212	11.743	6.2%	*
	r=1	0.029	1.276	30.3%	
United States	r=0	0.074	6.501	37.9%	

Summary of evidence for TB and NFA:

- Evidence of cointegration between TB and NFA for good number of variables.
- The number of rejection taking into account the size of the sample is quite good;

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Brazil:



TB and NFA:

Brazil TB, NFA

Rank	Log-likelihood	Trace Test		Max Test	
		[pvalue]		[pvalue]	
0	-38.60391	25.42 [0.008]	***	17.10 [0.008]	***
1	-30.05263	8.32 [0.073]	*	8.32 [0.073]	*
2	-25.89329				

***, **, * significance levels: 1%, 5% and 10%, respectively.

LRER and NFA:

System II LRER, NFA, LBSGDP and LTOT

Rank	Log-likelihood	Trace Test			Max Test		
		[pvalue]			[pvalue]		
		Trace test	[Prob]		Max test	[Prob]	
0	197,941	65,99	[0,002]	***	40,43	[0,000]	***
1	218,1563	25,56	[0,372]		11,36	[0,719]	
2	223,835	14,2	[0,282]		7,63	[0,601]	
3	227,6475	6,57	[0,156]		6,57	[0,155]	
4	230,9342						

Constant enters restricted in cointegrated space.

Model contains DI:2008 as unrestricted variable.

It is a dummy variable that has value 1 for 2008, -1 for 2009 and 0 otherwise.

LRER, NFA, TB and fundamentals:

Brazil System II LRER, NFA, LBSGDP and LTOT

Rank	Log-likelihood	Trace Test			Max Test		
		[pvalue]			[pvalue]		
		Trace test	[Prob]		Max test	[Prob]	
0	197,941	65,99	[0,002]	***	40,43	[0,000]	***
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Unrestricted Model: LRER, NFA and TB:

β	rer	tb	nfa	LBSGDP	LTOT	Constant
Vector 1	1	-0,84792	0,016691	-0,96894	-1,048	4,9596
Vector 2	-0,28459	1	0,051322	0,10602	0,43513	-0,99918

α	ΔRER	ΔTB	ΔNFA	$\Delta LBSGDP$	$\Delta LTOT$
Vector 1	-0,47396	-0,05763	-4,6163	-0,00586	0,38614
Vector 2	1,0853	0,14878	-14,955	0,12144	0,56097

Restricted Model: LRER, NFA and TB:

Model 1: Theoretical Identification

β	rer	nfa	tb	$\Delta LBSGDP$	$\Delta LTOT$	Constant
Vector 1	1	-1	0	-0,88896	-1	4,3126
Std error				0,09525		0,46158
Vector 2	0	3,2063	1	0	0	0
Std error		1,3494				
α	ΔRER	ΔNFA	ΔTB	$\Delta LBSGDP$	$\Delta LTOT$	
Vector 1	-0,66722	-0,0875	-2,8806	-0,02886	0,30609	
Std error	0,20301	0,09384	3,5379	0,050121	0,12778	
Vector 2	0,02238	0,008724	-0,44933	0,006246	0,022748	
Std error	0,010972	0,005072	0,19121	0,002709	0,006906	
LR Statistics	12.872	Distribution	$\chi^2(5)$		p-value	

* Likelihood Ratio Statistics is calculated by comparing unrestricted and Model i.

Restricted Model: LRER, NFA and TB:

- Cointegrated relationships:

$$ECM1_t = RER_t - NFA_t - 0.8896 * LBSGDP_t - LTOT_t + 4.3123$$

$$ECM2_t = TB_t + 0.032063 * NFA_t$$

- Misalignment equations:

$$misalignmentBRA_t = 1.1131 * ECM1_t - 0.091800 * ECM2_t$$

Restricted Model: LRER, NFA and TB:

- Cointegrated relationships:

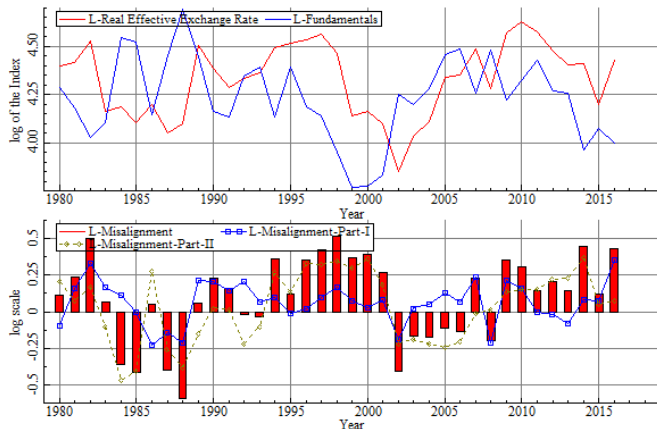
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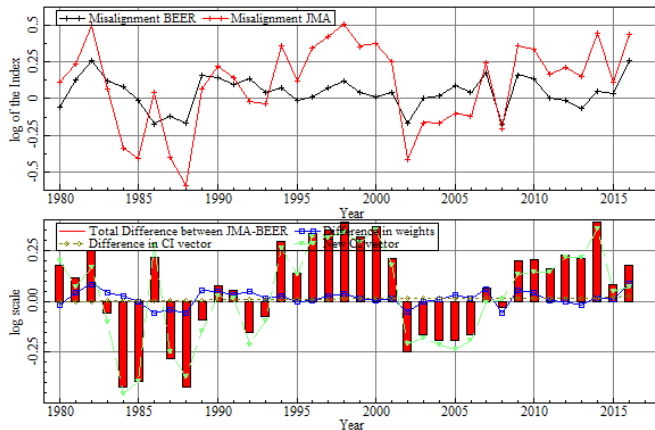
$$misalignmentBRA_t = 1.1131 * ECM1_t - 0.091800 * ECM2_t$$

Exchange Rate Misalignment Estimate:



L-Misalignment-Part-I is calculated from using ECM1.
 L-Misalignment-Part-II is calculated from using ECM2

What we learned from the model with TB:



Australia:



TB and NFA:

Australia

System III: TB, NFA

Rank	Log-likelihood	Trace Test	[p-value]		Max Test	[p-value]	
0	-18.36	19.95	[0.009]	***	19.06	[0.012]	**
1	-9.03	1.29	[0.257]		1.23	[0.268]	
2	-8.38						

Model is a VAR(2) with a dummy for 2008 and 2009.

Specification tests do not show evidence against the null of non auto-correlation, homoskedasticity and normality.

LRER, NFA and fundamentals:

Australia

System II: LRER, NFA, LTOT, LBSGDP

Rank	Log-likelihood	Trace Test	[p-value]		Max Test	[p-value]	
0	264.56	91.27	[0.000]	***	83.16	[0.000]	***
1	291.35	37.68	[0.004]	***	34.33	[0.013]	**
2	304.06	12.28	[0.145]		11.19	[0.203]	
3	310.03	0.33	[0.565]		0.3	[0.583]	
4	310.20						

Model is a VAR(2) with a dummy for 2008 and 2009.

Specification tests do not show evidence against the null of non auto-correlation, homoskedasticity and normality.

LRER, NFA, TB and fundamentals:

Australia

System I: LRER, NFA, TB, LTOT, LBSGDP

Rank	Log-likelihood	Trace Test	[p-value]		Max Test	[p-value]	
0	194.88	124.82	[0.000]	***	64.87	[0.000]	***
1	227.32	59.95	[0.002]	***	25.9	[0.079]	*
2	240.27	34.05	[0.014]	**	21.54	[0.042]	**
3	251.04	12.5	[0.135]		12.28	[0.100]	
4	257.18	0.22	[0.639]		0.22	[0.639]	
5	257.29						

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Restricted Model: LRER, NFA and TB:

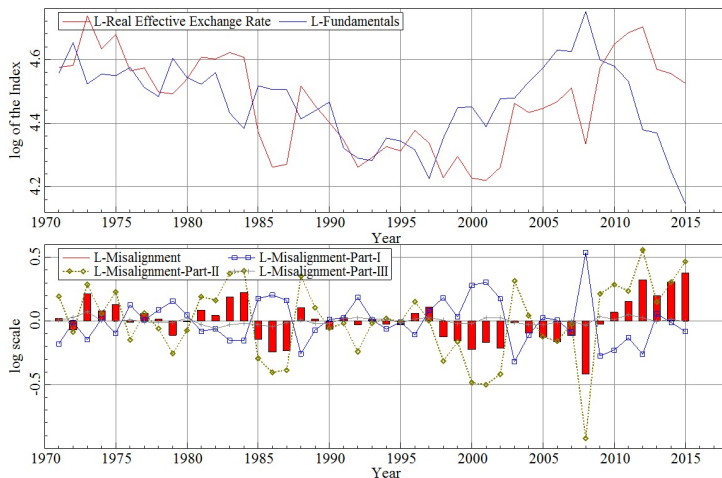
- Estimates cointegrated relationships:

$$ECM1_t = RER_t - 0.4279 * NFA_t - 0.30684 TOT_t$$

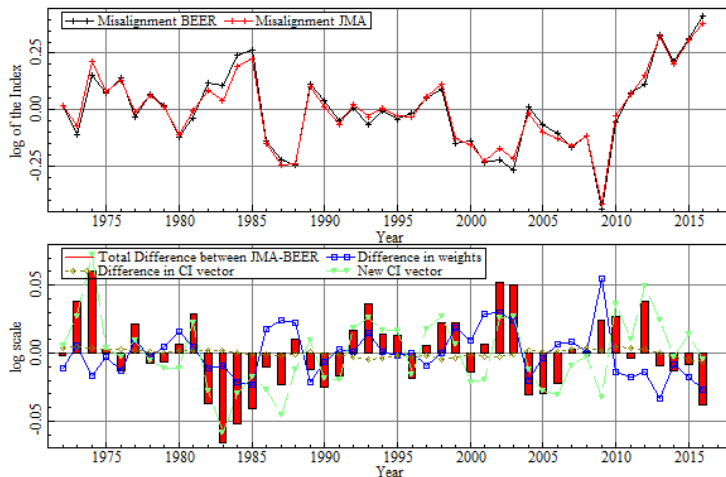
$$ECM2_t = RER_t - 0.39951 * LBSGDP_t - 0.34499 * TOT_t$$

$$ECM3_t = TB_t - 0.71287$$

Exchange Rate Misalignment Estimate:



What we learned from the model with TB:



New Zealand:



TB and NFA:

New Zealand

System III: TB, NFA

Rank	Log-likelihood	Trace Test	[p-value]		Max Test	[p-value]	
0	-58.89	34.21	[0.000]	***	30.49	[0.000]	**
1	-43.64	3.71	[0.469]		3.71	[0.469]	
2	-41.79						

Model is a VAR(2) .

Specification tests do not show evidence against the null of non auto-correlation, homoskedasticity and normality.

LRER, LTOT:

New Zealand

System II: LRER, NFA, LTOT, LBSGDP

Rank	Log-likelihood	Trace Test	[p-value]		Max Test	[p-value]	
0	99.53	20.73	[0.000]	***	18.45	[0.009]	***
1	108.75	2.28	[0.131]		2.28	[0.131]	
2	109.89						

Model is a VAR(2).

Specification tests do not show evidence against the null of non auto-correlation, homoskedasticity and normality.

LRER, NFA, TB and fundamentals:

New Zeland

System I: LRER, NFA, TB, LTOT

Rank	Log-likelihood	Trace Test	[p-value]		Max Test	[p-value]	
0	194.88	67.05	[0.002]	***	38.25	[0.001]	***
1	227.32	28.80	[0.21]		21.8	[0.057]	*
2	240.27	7.00	[0.89]		3.91	[0.95]	
3	251.04	3.08	[0.574]		3.08	[0.573]	
4	257.18						

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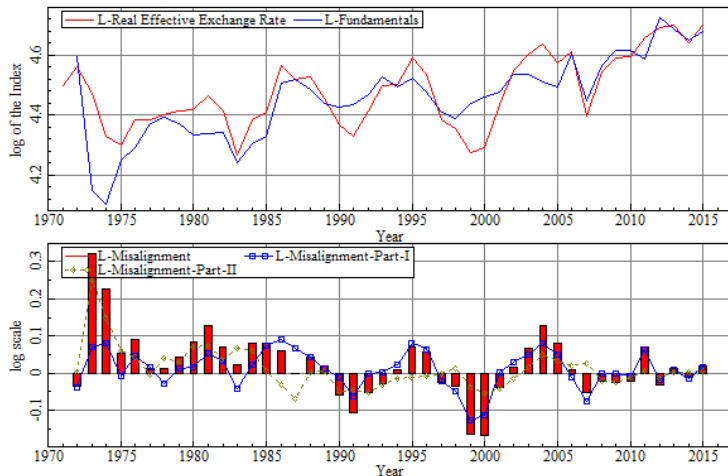
Restricted Model: LRER, NFA, TB and TOT

- Estimates of cointegrated relationships:

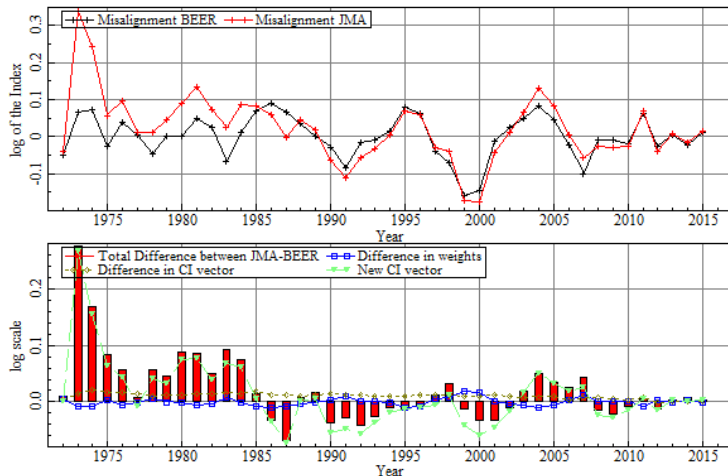
$$ECM1_t = RER_t - 1.0242 * TOT_t + 0.05651$$

$$ECM2_t = TB_t + 1.4479 * NFA_t$$

Exchange Rate Misalignment Estimate:



What we learned from the model with TB and NFA:



TB and NFA:

- For some countries TB may not be $I(1)$ if Absorption and GDP are cointegrated;
- But if TB is $I(0)$ then NFA should be $I(0)$ if discount rate is constant throughout time, assets and liabilities;
- Why discount rate should differ across time, assets and liabilities?

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Talking Points

- 1 Motivation and Goals
- 2 Approaches to address RER misalignment
- 3 Addressing Important Econometric Issues
- 4 Empirical Illustrations
- 5 Possible Extensions

Assets and Liabilities returns:

- Is there evidence of different returns on asset and liabilities;
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- GVAR seems to be a good option;
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